# Excretion in humans

Mark scheme and worked solutions · 20 questions

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## Question 1

core · 1 mark · 0610_T13_P1_20261013_Q1

**Answer: B**

**Rationale**
Carbon dioxide is excreted through the lungs: it diffuses from the blood into the alveoli and is breathed out. The kidneys excrete urea, excess water and excess ions, not carbon dioxide. The liver makes urea but does not excrete carbon dioxide. The stomach is part of the digestive system and is not an excretory organ.

## Question 2

core · 1 mark · 0610_T13_P1_20261013_Q2

**Answer: D**

**Rationale**
Carbon dioxide is excreted through the lungs and urea is excreted by the kidneys. Option A swaps the two organs. Options B and C name the liver, which forms urea but does not excrete either substance; making a waste substance is not the same as excreting it.

## Question 3

core · 1 mark · 0610_T13_P1_20261013_Q3

**Answer: A**

**Rationale**
The kidneys excrete urea, excess water and excess ions in the urine. Option B is wrong because glucose is kept in the body and carbon dioxide is excreted through the lungs. Option C is wrong because carbon dioxide leaves through the lungs and proteins are not excreted. Option D is wrong because glucose and amino acids are useful nutrients that the body keeps.

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## Question 4

core · 1 mark · 0610_T13_P1_20261013_Q4

**Answer: C**

**Rationale**
Drinking a large volume of water gives the blood more water than the body needs. The kidneys excrete this excess water, so more urine is produced. Option A is wrong because water is not converted into urea; urea is made from excess amino acids. Option B is wrong because the kidneys continue to excrete urea, and this would not increase the volume of urine. Option D is wrong because the bladder stores urine; it does not absorb water, and absorbing water would reduce the volume of urine.

## Question 5

supplement · 1 mark · 0610_T13_P1_20261013_Q5

**Answer: A**

**Rationale**
Deamination is the removal of the nitrogen-containing part of amino acids to form urea. Option B describes the use of amino acids to make proteins, which is assimilation. Option C describes the digestion of proteins. Option D describes filtration in the kidney, which removes urea from the blood but does not make it.

## Question 6

supplement · 1 mark · 0610_T13_P1_20261013_Q6

**Answer: D**

**Rationale**
In a healthy person, most of the water in the filtrate is reabsorbed into the blood; the rest leaves in the urine. Option A is wrong because some water must stay in the nephron to form urine. Option B is wrong because the body would lose a huge volume of water each day. Option C is wrong because the glomerulus is where substances are filtered out of the blood; reabsorption happens as the filtrate flows along the rest of the nephron.

## Question 7

supplement · 1 mark · 0610_T13_P1_20261013_Q7

**Answer: B**

**Rationale**
Using absorbed amino acids to build proteins is assimilation: the liver converts amino acids into proteins. Option A, deamination, removes the nitrogen-containing part of excess amino acids to form urea and does not make proteins. Option C, digestion, breaks large molecules down into small ones and happens in the alimentary canal. Option D, egestion, is the passing out of undigested food as faeces.

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## Question 8

supplement · 1 mark · 0610_T13_P1_20261013_Q8

**Answer: C**

**Rationale**
Amino acids cannot be stored in the body. Excess amino acids are deaminated in the liver: the nitrogen-containing part is removed and urea is formed. Option A is wrong because the kidneys do not store amino acids. Option B is wrong because the amino acids have already been absorbed into the blood, so they cannot be egested in faeces. Option D is wrong because the lungs excrete carbon dioxide, not amino acids.

## Question 9

supplement · 1 mark · 0610_T13_P1_20261013_Q9

**Answer: D**

**Rationale**
Glucose is filtered at the glomerulus, and normally all of it is reabsorbed into the blood as the filtrate flows along the nephron. When the blood glucose concentration is very high, the filtrate contains so much glucose that the nephrons cannot reabsorb all of it, so some stays in the urine. Option A is wrong because urea is not converted into glucose. Option B is wrong because glucose is filtered at the glomerulus, and if it were not filtered it could not appear in the urine. Option C is wrong because there is no exchange of urea for glucose.

## Question 10

supplement · 1 mark · 0610_T13_P1_20261013_Q10

**Answer: A**

**Rationale**
Urea is toxic, so if it were not excreted it would build up in the blood and harm the body. Option B is wrong because urea is a waste product and is not used as a source of energy. Option C is wrong because urea is not converted back into amino acids. Option D is wrong because proteins are made from amino acids, not from urea.

## Question 11

core · 5 marks · 0610_T13_P1_20261013_Q11

**(a)** [1]

- lungs (1)
- ALLOW: lung
- REJECT: mouth; nose; trachea

**(b)** [1]

- (aerobic) respiration (1)
- REJECT: breathing; gas exchange; photosynthesis

**(c)** [3]

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- urea (1)
- (excess) water (1)
- (excess) ions (1)
- ALLOW: (excess) salts / named ions (e.g. sodium ions) for ions
- REJECT: glucose; carbon dioxide; protein; amino acids

**Worked solution**
(a) Carbon dioxide diffuses from the blood into the air in the lungs and is breathed out, so the lungs excrete it. (b) Carbon dioxide is a waste product of respiration, which happens in all living cells. Breathing is the movement of air into and out of the lungs; it is not the process that makes carbon dioxide. (c) The kidneys excrete urea, excess water and excess ions in the urine. Glucose is not excreted because the body needs it, and carbon dioxide leaves through the lungs, not the kidneys.

## Question 12

core · 6 marks · 0610_T13_P1_20261013_Q12

**(a)** [1]

- drinking more liquid increases the volume of urine produced (1)
- ALLOW: an answer using the data, e.g. drinking 1000 cm³ more liquid increases urine by 900 cm³

**(b)** [2]

- increase = 2100 − 1200 = 900 (cm³) and 900 ÷ 1200 × 100 (1)
- 75 (%) (1)
- ALLOW: 75 (%) with no working for 2 marks

**(c)** [2]

- she loses (more) water in sweat (when running on a hot afternoon) (1)
- ALLOW: loses more water in breath / by evaporation from the skin
- so there is less excess water in the blood (for the kidneys to excrete) / the kidneys excrete less water (1)

**(d)** [1]

- urea (1)
- REJECT: urine; uric acid; ammonia; nitrogen

**Worked solution**
(a) On day 2 she drinks 1000 cm³ more than on day 1 and produces 900 cm³ more urine, so drinking more liquid increases the volume of urine. (b) Percentage increase = (new value − old value) ÷ old value × 100 = (2100 − 1200) ÷ 1200 × 100 = 900 ÷ 1200 × 100 = 75%. Check: 1200 × 1.75 = 2100. (c) During a long run on a hot afternoon she sweats a lot and loses water from her body. This means there is less water in excess in her blood, so the kidneys excrete less water and the volume of urine is smaller. (d) The kidneys excrete urea, which is the nitrogen-containing waste substance in urine.

## Question 13

core · 5 marks · 0610_T13_P1_20261013_Q13

**(a)** [2]

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- P is a lung because carbon dioxide (concentration) decreases (and urea does not change) (1)
- Q is a kidney because urea (concentration) decreases (1)
- no mark for naming the organ without a reason from the data

**(b)** [2]

- (cells in) organ Q respire (1)
- carbon dioxide (made in respiration) passes / diffuses into the blood (1)
- ALLOW: kidneys do not excrete carbon dioxide (so it is not removed from the blood) for the second mark

**(c)** [1]

- the lungs do not excrete urea / urea is excreted by the kidneys (not the lungs) (1)
- ALLOW: the lungs only excrete carbon dioxide

**Worked solution**
(a) The lungs excrete carbon dioxide, so blood leaving a lung has less carbon dioxide than blood entering it. That happens in P (52 to 48), so P is a lung. The kidneys excrete urea, so blood leaving a kidney has less urea. That happens in Q (0.30 to 0.24), so Q is a kidney. (b) Kidney cells are living cells and they respire. Respiration makes carbon dioxide, which passes into the blood. The kidneys do not excrete carbon dioxide, so the concentration rises. (c) Urea is excreted by the kidneys. The lungs do not remove urea, so its concentration stays the same in P.

## Question 14

supplement · 5 marks · 0610_T13_P1_20261013_Q14

**(a)** [1]

- (the liver) converts amino acids into proteins (1)
- ALLOW: uses amino acids to make proteins
- REJECT: stores amino acids; breaks down proteins

**(b)** [2]

- removal of the nitrogen-containing part (of amino acids) (1)
- (of excess) amino acids to form urea (1)
- ALLOW: removal of the amine group / $NH_2$ group for the first mark

**(c)** [1]

- liver (1)

**(d)** [1]

- kidneys (1)
- ALLOW: kidney

**Worked solution**
(a) In assimilation, the liver converts amino acids into proteins, which the body uses. (b) Amino acids cannot be stored, so excess amino acids are deaminated. Deamination is the removal of the nitrogen-

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containing part of amino acids to form urea. (c) Deamination happens in the liver, so urea is formed in the liver. (d) Urea passes into the blood and is carried to the kidneys, which excrete it in the urine.

## Question 15

supplement · 5 marks · 0610_T13_P1_20261013_Q15

**(a)** [1]

- filtration (1)
- ALLOW: (ultra)filtration
- REJECT: diffusion; osmosis; reabsorption

**(b)** [2]

- water, glucose, urea, ions
- all four correct (2)
- two or three correct (1)
- ALLOW: salts / named ions (e.g. sodium ions) for ions
- REJECT: red blood cells; protein (deduct one mark for each wrong substance given, minimum 0)

**(c)** [2]

- all of the glucose (1)
- is reabsorbed into the blood (as the filtrate flows along the nephron) (1)
- REJECT: glucose is not filtered; glucose is used up in the nephron

**Worked solution**
(a) Substances are filtered out of the blood in the glomerulus; this is filtration. (b) The substances filtered out of the blood into the nephron are water, glucose, urea and ions. (c) Glucose is filtered at the glomerulus, so it is in the filtrate. As the filtrate flows along the nephron, all of the glucose is reabsorbed into the blood, so in a healthy person no glucose is left in the urine. The body keeps glucose because it is needed for respiration.

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## Question 16

supplement · 6 marks · 0610_T13_P1_20261013_Q16

**(a)** [1]

- the concentration of urea is the same in the blood plasma and the filtrate (1)
- ALLOW: urea is present in the filtrate at 0.3 (g per $dm^3$) like the plasma

**(b)** [1]

- 60 (times) (1)

**(c)** [2]

- most of the water (in the filtrate) is reabsorbed into the blood (1)
- urea stays in the nephron / (little or) no urea is reabsorbed / much less urea is reabsorbed than water (1)
- ALLOW: water is reabsorbed but urea is not (for 2 marks)
- REJECT: urea is made / added in the nephron

**(d)** [2]

- water reabsorbed = 180 − 1.5 = 178.5 ($dm^3$) and 178.5 ÷ 180 × 100 (1)
- ALLOW: 1.5 ÷ 180 × 100 = 0.833 (%) not reabsorbed, then 100 − 0.833, for the method mark
- 99.2 (%) (1)
- ALLOW: 99.17 (%) for 2 marks
- 99 (%) scores 1 mark only (the question asks for three significant figures)

**Worked solution**
(a) Plasma entering the glomerulus and filtrate just after it both contain urea at 0.3 g per $dm^3$, so urea has passed out of the blood into the nephron: it has been filtered. (b) 18.0 ÷ 0.3 = 60, so urea is 60 times more concentrated in urine than in filtrate. Check: 0.3 × 60 = 18.0. (c) As the filtrate flows along the nephron, most of the water is reabsorbed into the blood, but the urea stays in the nephron. The same urea is dissolved in much less water, so it becomes more concentrated. (d) The water that is not reabsorbed leaves as urine, so the volume reabsorbed is 180 − 1.5 = 178.5 $dm^3$. Percentage reabsorbed = 178.5 ÷ 180 × 100 = 99.166…% = 99.2% to three significant figures. Check another way: 1.5 ÷ 180 × 100 = 0.833% leaves as urine, and 100 − 0.833 = 99.2%.

## Question 17

supplement · 6 marks · 0610_T13_P1_20261013_Q17

**(a)** [2]

- as the mass of protein eaten increases, the mass of urea in the urine increases (1)
- the mass of urea is (about) proportional to the mass of protein / doubling the protein (about) doubles the urea / urea increases by about 12–13 g for every 40 g of protein (1)
- ALLOW: a correct quoted pair of data, e.g. 40 g protein gives 12 g urea and 80 g protein gives 25 g urea, for the second mark

**(b)** [3]

- more protein is digested so more amino acids are absorbed (into the blood) (1)
- more amino acids are in excess / more amino acids than are needed to make proteins (1)
- (excess amino acids are) deaminated in the liver / the nitrogen-containing part is removed in the liver (1)
- (to form) urea (1)
- max 3

**(c)** [1]

- urea is toxic (1)
- ALLOW: urea would build up (in the blood) and poison / harm cells
- REJECT: urea takes up space; urea is not needed (without reference to toxicity)

**Worked solution**
(a) Read the table: each time the protein eaten goes up by 40 g, the urea in the urine goes up by about 12–13 g. So more protein gives more urea, and doubling the protein (40 g to 80 g, or 80 g to 160 g) roughly doubles the urea. (b) Protein is digested into amino acids, which are absorbed into the blood. The liver uses some amino acids to make proteins, but amino acids cannot be stored. When more protein is eaten, there are more excess amino acids. The liver deaminates them: it removes the nitrogen-containing

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part, which forms urea. The kidneys then excrete the urea, so more appears in the urine. (c) Urea is toxic. If it stayed in the body it would build up in the blood and harm the body, so it must be excreted.

## Question 18

supplement · 6 marks · 0610_T13_P1_20261013_Q18

**(a)** [2]

- decrease = 8.0 − 3.0 = 5.0 and 5.0 ÷ 8.0 × 100 (1)
- 62.5 (%) (1)
- ALLOW: 63 (%) or 62 (%)
- ALLOW: 62.5 (%) with no working for 2 marks

**(b)** [2]

- (the liver) converts amino acids into proteins / assimilation (1)
- (excess) amino acids are deaminated (1)
- ALLOW: (excess) amino acids are converted into urea for the second mark

**(c)** [2]

- (excess) amino acids are deaminated in the liver / the nitrogen-containing part of amino acids is removed (1)
- (which forms) urea, which passes into the blood (leaving the liver) (1)

**Worked solution**
(a) Percentage decrease = (starting value − final value) ÷ starting value × 100 = (8.0 − 3.0) ÷ 8.0 × 100 = 5.0 ÷ 8.0 × 100 = 62.5%. Check: 8.0 × (1 − 0.625) = 3.0. (b) The liver takes amino acids out of the blood in two ways. It converts some into proteins (assimilation). Excess amino acids cannot be stored, so the liver deaminates them. (c) In deamination the nitrogen-containing part of each amino acid is removed and urea is formed. The liver releases this urea into the blood, so blood leaving the liver contains more urea than blood arriving from the small intestine. The urea is then carried to the kidneys to be excreted.

## Question 19

supplement · 6 marks · 0610_T13_P1_20261013_Q19

**(a)** [3]

- statement 2 (1)
- statement 4 (1)
- statement 5 (1)
- deduct one mark for each correct statement (1, 3 or 6) given, minimum 0

**(b)** [3]

- statement 2: urea is formed in the liver (from excess amino acids) (1)
- statement 4: some of the ions in the filtrate are reabsorbed into the blood (1)
- statement 5: most of the water in the filtrate is reabsorbed into the blood (1)
- a correction mark can only be given for a statement correctly given in (a)
- REJECT: statement 4 corrected to "all of the ions are reabsorbed"; statement 5 corrected to "all of the water is reabsorbed"; statement 2 corrected only to "urea is excreted by the kidneys"

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**Worked solution**
Check each statement against what you know. Statement 1 is correct: carbon dioxide is excreted through the lungs. Statement 2 is wrong: urea is formed in the liver by deamination of excess amino acids; the kidneys only excrete it. Statement 3 is correct: after the useful substances and most of the water are reabsorbed, the liquid left at the end of the nephron is urine, containing urea, excess water and excess ions. Statement 4 is wrong: some of the ions are reabsorbed into the blood, and the excess ions leave in the urine. Statement 5 is wrong: most of the water is reabsorbed into the blood; only a small amount leaves as urine. Statement 6 is correct: urea is toxic, so it must be excreted.

## Question 20

supplement · 7 marks · 0610_T13_P1_20261013_Q20

**(a)** [2]

- 125 × 60 = 7500 ($cm^3$) (1)
- 7.5 ($dm^3$) (1)
- ALLOW: 7500 $cm^3$ for 1 mark if not converted to $dm^3$

**(b)** [1]

- 7.5 × 0.9 = 6.75 g (1)
- ALLOW: 6.8 g
- ECF: answer to (a) in $dm^3$ × 0.9, with unit g

**(c)** [2]

- decrease = 125 − 30 = 95 ($cm^3$ per minute) and 95 ÷ 125 × 100 (1)
- 76 (%) (1)
- ALLOW: 76 (%) with no working for 2 marks

**(d)** [2]

- less blood is filtered / less urea is filtered out of the blood (at the glomeruli) (1)
- so less urea is excreted / removed in the urine, but the liver keeps forming urea (from excess amino acids), so urea builds up in the blood (1)

**Worked solution**
(a) There are 60 minutes in an hour, so the volume of filtrate is 125 × 60 = 7500 $cm^3$. There are 1000 $cm^3$ in 1 $dm^3$, so this is 7500 ÷ 1000 = 7.5 $dm^3$. Check: 7.5 $dm^3$ ÷ 60 = 0.125 $dm^3$ = 125 $cm^3$ per minute. (b) Mass = volume × concentration = 7.5 $dm^3$ × 0.9 g per $dm^3$ = 6.75 g. Check: 6.75 ÷ 7.5 = 0.9. (c) Percentage decrease = (125 − 30) ÷ 125 × 100 = 95 ÷ 125 × 100 = 76%. Check: 125 × 0.24 = 30. (d) Urea is removed from the blood when it is filtered at the glomeruli and then excreted in the urine. The damaged glomeruli filter much less blood, so much less urea is removed. The liver keeps forming urea from excess amino acids, so urea builds up in the blood.

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